

KP_01 3D LiDAR Evaluation Kit User Guide

Get Started

Prerequisites

- Power supply capable of outputting at least 12V (16.8V recommended) at minimum 0.3A
- Computer running 3D Serial Plotter using Python

Set up

1. Plug in the three plugs from the evaluation module into the LiDAR (Make sure the black plug is plugged into the black socket)
2. Connect alligator clips from the power supply to the power leads on the evaluation module
3. Set the voltage of the power supply to 16.8V (or anything above 12V, if applicable)
4. Turn on the power supply. The lidar will make a startup sound
5. Plug in the USB-C cable into the computer, and select the correct port in SerialStudio
6. Hold the base of the LiDAR securely until it slows down
7. Click “connect” in SerialStudio. A 3D plot will appear with measured points

Usage

Function Description

The KP_01 3D LiDAR measures 3D space around it by firing an infrared laser and timing its return. The LiDAR pans continuously using its motor, and tilts 50 degrees downwards and upwards for a 100 degree vertical FoV. The LiDAR itself is accurate to 1cm, and has a maximum range of 8m. Points beyond that will automatically be snapped to 8m away.

Emphasis Modes

The KP_01 3D LiDAR supports four different emphasis modes. These limit the tilt servo sweep range and can be used for real-time obstacle avoidance with the benefit of frequent updates and therefore higher point density.

These are the different emphasis modes:

- Default (no emphasis):
 - Enabled by default. The tilt servo does a full sweep of its 100 degree range. Enable by typing “clear” in the Serial port
- Bottom emphasis:
 - The tilt servo only sweeps across the bottom 20 degrees of its range. Enable by typing “bottom” in the Serial port
- Middle emphasis:
 - The tilt servo only sweeps across the middle 20 degrees of its range. Enable by typing “middle” in the Serial port
- Top emphasis:
 - The tilt servo only sweeps across the top 20 degrees of its range. Enable by typing “top” in the Serial port

Low Latency Mode

The KP_01 3D LiDAR can also increase its revolutions per second (RPS) for a much faster update time suitable for real-time obstacle avoidance again. However, this compromises angular resolution due to the LiDAR itself not being able to update any faster. This isn't an issue in scenarios where fast update time is prioritised.

Enable low latency mode by typing “llm” in the Serial port, and disable it by typing “llmOff” in the Serial port.